

- 2 A program reads data from a text file, splits the data depending on its content and stores the separated data into different files.

The text file `TheData.txt` contains 72 lines of data. Each line of data has an integer number and a string colour that are separated by a comma. For example, the first line in the file is:

```
10,red
```

The integer is 10 and the string is "red"

The file contains six different colours: red, green, blue, orange, yellow, pink.

(a) The function `ReadData()`:

- prompts the user to enter a filename and reads this filename from the user
- opens the file and reads each line of data into a 1D array
- returns the populated 1D array.

The function needs to work for a file that contains an unknown number of lines.

Write program code for `ReadData()`

Save your program as **Question2_J25**.

Copy and paste the program code into part **2(a)** in the evidence document.

[7]

(b) The procedure `SplitData()` takes a 1D string array as a parameter with the identifier `DataArray`

The procedure declares six 1D arrays: one array for each colour that appears in the file (red, green, blue, orange, yellow, pink).

The procedure accesses each string in `DataArray`. The data in each string is split into the integer and the colour. The integer is stored in the array that matches the colour.

For example, the first string in `DataArray` has the integer 10 and the colour red, so the integer 10 is stored in the array for the colour red.

Write program code for `SplitData()`

Save your program.

Copy and paste the program code into part **2(b)** in the evidence document.

[6]